M^{3}FuNet: An Unsupervised Multivariate Feature Fusion Network for Hyperspectral Image Classification

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***Abstract*— Hyperspectral image (HSI) spectral-spatial joint feature (FE) extraction methods generally suffer from low feature retention and weak spatial–spectral dependence, which will lead to single-class feature confrontation (SCFC). To solve this problem, an unsupervised multivariate feature fusion network (M^{3}FuNet) is developed in this article. In M^{3}FuNet, multiscale supervector matrix correction (MSMC) and multiscale random convolution dispersion (MRCD) are used as the spectral and spatial feature extraction method, and the feature retention of spectral and spatial features is improved to achieve feature calibration by feature fusion and decision fusion, called "multivariate feature fusion." The MSMC is employed to correct the supervector matrix to reduce the intraclass variance in superpixel homogeneous regions and overcome the phenomenon of supervector block drift (SvBD). The MRCD uses random convolution and Gaussian smoothing to extract deep spatial features. Because of the similar feature representation ability of the MSMC and MRCD, the obtained spectral-spatial joint features have high feature retention and strong spectral-spatial dependence. Finally, this M^{3}FuNet is used for realizing the classification of HSI. Three common HSI datasets are used to validate the effectiveness of the M^{3}FuNet. The experiment results show that the M^{3}FuNet has a superior performance compared with several state-of-the-art (SOTA) HSI classification methods. The code of the proposed M^{3}FuNet is available at http://github.com/aichou233/M3FuNet.**

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***Index Terms*— Gaussian smoothing, hyperspectral image classification (HSIC), multivariate feature (FE) fusion, supervector matrix correction (SMC).**

I. Introduction

HYPERSPECTRAL images (HSI) have high feature (FE) dimensions and rich and continuous spectral curves of features, which have the characteristics of "image spectral in one" [1]. HSI classification (HSIC) is used extensively in city planning, agricultural survey, national defense security, ecological environment detection, and other fields [2], [3], [4]. In the practical application of HSIC, there is a lot of superfluous information in the high feature dimensions, and with the increase of feature dimensions, the HSIC accuracy will appear in the Hughes phenomenon of "increase and then decrease" [5].

Traditional HSIC generally learns pixel-level features directly, and some joint sparse representation (JSR) [6] models are able to learn the important weights of neighboring pixels, such as maximum likelihood estimation (MLE)-based JSR (MLEJSR) [7] significantly reduces the influence of inhomogeneous neighboring pixel values. Self-paced JSR (SPJSR) [8] gradually adds neighboring pixels from easy to complex to the JSR learning process through an alternating iteration strategy, which ultimately generates an ideal model. In addition to the direct pixel-level classification, dimensionality reduction is also an important preprocessing method. Traditional unsupervised dimensionality reduction methods are principal component analysis (PCA) [9], [10], and independent component analysis (ICA) [11], but they are difficult to model nonlinear relationships in complex data. Superpixel PCA (SuperPCA) [12], a dimensionality reduction method based on entropy rate superpixel (ERS) [13], will capture the complex spectral features of the superpixel homogeneous regions. But a phenomenon of supervector block drift (SvBD) will constrain the spectral representation ability of SuperPCA. SvBD is the manifestation of spectral variability [14] of HSI (same class but different spectral) on superpixel in Fig. 1(a). All the bands of a particular superpixel are stretched into vectors and stitched together to obtain the supervector in Fig. 1(b). Some bands (supervector block) can deviate significantly

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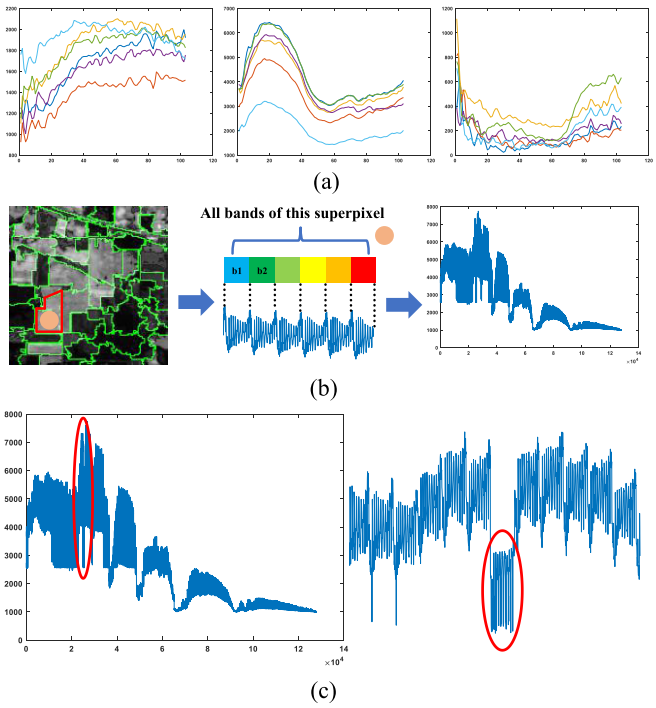


Fig. 1. (a) Spectral curves of the same features. (b) Formation of supervector. (c) SvBD of supervector.

from the local variation of the supervector, and destroy the smoothness of the supervector in Fig. 1(c). On the other hand, deep learning methods can handle nonlinear data better and outperform traditional machine learning-based feature extraction methods [15], [16], [17]. Convolutional neural networks (CNNs) [18], [19] extract effective local spatial features from HSI. Generative adversarial networks (GANs) [20] mine deep spatial and spectral domain correlations. Recurrent neural networks (RNNs) [21] process spectral curves as sequence data to extract global spectral feature. Hong et al. [22] investigated graph convolution neural networks (GCNs) to capture short-medium-long range spatial relations in HSI. Hong et al. [23] proposed a spectral transformer (SpectralFormer) to learn the spectral local sequence features in HSI. Xu et al. [24] proposed a random patch network (RPNet), which extracts random patches as convolution kernels from the HSI with high classification efficiency, but the problem of unstable representation can arise. Although these deep learning methods can extract the deep features of HSI, they tend to ignore the spectral-spatial joint feature information.

Spectral features and spatial features tend to retain different feature information, and spatial–spectral joint features are often used to improve classification accuracy [25], [26]. Fu et al. [27] proposed a tensor singular spectral analysis (TensorSSA) to extract global and low-rank 3-D features. Lu et al. [28] proposed a spectral-spatial and superpixel-wise unsupervised linear discriminant analysis (S^3-ULDA) to extract global spatial–spectral joint features and local spatial–spectral joint features. Zhong et al. [29] proposed a spectral-spatial transformer network (SSTN) to overcome the constraints of convolution kernels. Roy et al. [30] proposed a spectral-spatial morphological attention transformer (morphFormer) to implements a learnable spectral and spatial morphological network. Zhou et al. [31] proposed a masked spectral-spatial feature prediction (MSSFP) to help the transformer understand the complicated spectral-spatial structures of unlabeled HSI data. These spectral-spatial joint feature methods have been a great success at HSIC, but there are still aspects that lack consideration. The spectral-spatial dependence of their extracted joint features is weak, and they rely on the representation ability of the joint features in the classification module, which tends to cause single-class feature confrontation (SCFC). Because of the excessive difference between spectral representation (the representation ability of spectral features) and spatial representation (the representation ability of spatial features), the joint representation (the representation ability of spectral-spatial joint features) tends to be in between the spectral and spatial representation. From a single-class perspective, the classification accuracy of single class can be significantly lower than that of the same class on spectral and spatial features.

In order to alleviate the mentioned SvBD and SCFC in spectral-spatial joint feature classification of HSI and to alleviate the unstable representation of random convolution features, an unsupervised multivariate feature fusion classification network (M^3FuNet) is developed in this article. To alleviate the SvBD, a supervector matrix correction (SMC) method is proposed to correct the supervector matrix to reduce the intraclass variance in superpixel homogeneous regions (subregions), and the interpretability of the spectral curve in subregions can be enhanced. Then, a new spectral feature extraction method based on multiscale SMC (MSMC) is proposed, which uses SMC and SuperPCA to capture the spectral features in multiscale subregions. To alleviate the unstable representation of random convolution features, random convolution and Gaussian smoothing are integrated to propose a new spatial feature extraction method based on multiscale random convolution dispersion (MRCD) for extracting deep spatial features in a multiscale structure. To alleviate the SCFC, spectral information or spatial information is not considered singly in the MSMC and MRCD methods as much as possible so that the feature retention and spectral-spatial dependence of joint features are improved by making the extracted features have similar representation ability. The extracted MSMC and MRCD features are fused to obtain joint features, applying decision fusion [32], [33] to spectral, spatial, and joint features for feature calibration (this process is called "multivariate feature fusion").

The main contributions are summarized as follows.

1) An unsupervised multivariate feature fusion network (M^3FuNet) is developed to mine multiscale, deep spectral and spatial features of HSI, and fuse spectral and spatial features with similar representation ability, and employ decision fusion to achieve feature calibration.
2) A spectral feature extraction method based on MSMC is proposed. which uses the SMC to correct the supervector matrix to reduce the intraclass variance in multiscale subregions, and further extracts multiscale spectral features.

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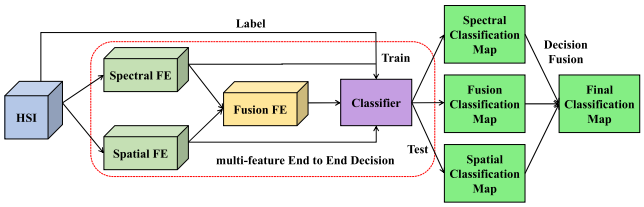


Fig. 2. Multivariate feature fusion framework.

3) A spatial feature extraction method based on MRCD is proposed, which introduces Gaussian smoothing to enhance the smoothness of the random convolution feature edges and improve the robustness of the random convolution features, and further extract deep spatial features.

The rest of the article is organized as follows. Section II describes the framework of M^{3}FuNet. Section III describes the MSMC in detail. Section IV describes the MRCD in detail. Experimental results and analyses are given in Section V. Section VI outlines the conclusions.

Explanation of basic symbols: this article uses italicized letters for scalars (e.g., x and X), bold lowercase letters to represent vectors (e.g., $\mathbf{x}$), bold uppercase letters to represent matrices (e.g., $\mathbf{X}$), and art letters to represent a tensor (e.g., χ). Using $\mathbb{R}$ to represent the real number field.

II. Multivariate Feature Fusion Network

M^{3}FuNet is an end-to-end unsupervised multivariate feature fusion classification network, and the framework of the network is shown in Fig. 2. M^{3}FuNet extracts spectral FE and spatial FE from HSI (the HSI raw data is defined as $\chi = [\mathbf{P}_1, \mathbf{P}_2, \cdots \mathbf{P}_B] \in \mathbb{R}^{H\times W\times B}$, $\mathbf{P}_i \in \mathbb{R}^{H\times W}$ denotes the ith spectral band of HSI, H denotes the length of HSI, W denotes the width of HSI, the number of bands is B), and then obtains spatial–spectral joint features (fusion FE), and then inputs the spectral FE, spatial FE, and fusion FE into a classifier in parallel for training, and finally obtains spectral classification map, spatial classification map and fusion classification map, respectively. The final classification map is obtained by decision fusion.

The specific flow of M^{3}FuNet is shown in Fig. 3, in which the spectral feature extraction method MSMC method is based on the superpixel guidance to extract the spectral features $\chi_{\text{spectral}} \in \mathbb{R}^{H\times W\times B_1}$. The spatial feature extraction method MRCD is based on the deep random patches convolution (RPC) to extract the spatial features $\chi_{\text{spatial}} \in \mathbb{R}^{H\times W\times B_2}$. M^{3}FuNet fuses the χ_{spectral} and χ_{spatial} to obtain spatial–spectral joint features $\chi_{\text{joint}} \in \mathbb{R}^{H\times W\times B_3}$. Joint features have the advantage of complementary spectral-spatial representation due to the difference of the extracted features by MSMC and MRCD.

Because of the dimensions of the extracted different features are inconsistent, the efficient concatenation fusion is adopted here, which has the advantage of high feature retention

$$\chi_{\text{joint}} = \left[\chi_{\text{spectral}},\ \chi_{\text{spatial}}\right] \in \mathbb{R}^{H\times W\times B_3},\ B_3 = B_1 + B_2. \quad (1)$$

Before inputting to the classifier, to avoid the problem of singular values and gradient explosion, χ_{joint} is further converted to a 2-D matrix $\mathbf{X}_{\text{joint}} \in \mathbb{R}^{L\times B_3}(L = H \times W)$ for normalization operation

$$\mathbf{X}^*_{\text{joint}} = \frac{\mathbf{X}_{\text{joint}} - \mathbf{X}_{\text{mean}}}{\mathbf{X}_{\text{std}}} \quad (2)$$

where $\mathbf{X}_{\text{mean}}$ is the mean of $\mathbf{X}_{\text{joint}}$, $\mathbf{X}_{\text{std}}$ is the standard deviation of $\mathbf{X}_{\text{joint}}$.

M^{3}FuNet will retain the feature contribution of MSMC and MRCD and input the χ_{spectral}, χ_{spatial} and χ_{joint} in parallel into classifier to obtain the spectral decision $R_1 \in \mathbb{R}^{H\times W}$, spatial decision $R_2 \in \mathbb{R}^{H\times W}$ and the joint decision $R_3 \in \mathbb{R}^{H\times W}$, respectively, and then obtain the final classification decision $R_f \in \mathbb{R}^{H\times W}$ by decision fusion. This process achieves feature calibration by feature fusion and decision fusion, which is called "multivariate feature fusion" in this article. The decision fusion results can be obtained

$$R_i = \text{class}(x, y)_i = \text{elemax}\sum\nolimits_{j=1}^{B} i\chi_c^{B_i}(x, y, j) \quad (3)$$

$$R_f = \text{elemax}\sum\nolimits_{i=1}^{3} \gamma_i Ri(x, y) \quad (4)$$

where elemax$\sum(\cdot)$ denotes the element that occurs most in the summed polynomials, γ_i represents the decision strength of the ith features. Here the value of γ_i is constantly set to 1.

III. Multiscale Supervector Matrix Correction

A. Superpixel Principal Component Analysis

To extract the strong spectral-spatial dependence, the ERS [13] algorithm is used to generate subregions (superpixel) in the HSI that do not overlap each other. The graph $\mathbf{IMG} = (\mathbf{V}, \mathbf{A})$ is divided into many subregions by selecting edges $A \subseteq E$, and the entropy rate term $H(\mathbf{A})$ and the balance term $F(\mathbf{A})$ are introduced in the objective function of ERS to optimize the segmentation

$$A^* = \text{argmax}(\text{tr}(H(A) + \gamma F(\mathbf{A}))), \quad A \subseteq E \quad (5)$$

where tr$(\cdot)$ is trace function, $\gamma > 0$ is used to balance the impact level of $H(\mathbf{A})$ and $F(\mathbf{A})$.

For ease of handling, the HSI cube data $\chi \in \mathbb{R}^{H\times W\times B}$ is reconstructed into a 2-D matrix such that $\mathbf{X} \in \mathbb{R}^{L\times B}(L = M \times N)$. PCA finds a linear mapping from the $\mathbf{X}$, specifies the target dimension $b(b < B)$, which completes the mapping of B to b dimension ($\mathbf{Y} = \mathbf{Q}^T\mathbf{X}$). The projection matrix $\mathbf{Q}$ is obtained by following objective function:

$$\mathbf{Q}^* = \text{argmax}\left(\text{tr}\left(\mathbf{Q}^T\text{Cov}(\mathbf{X})\mathbf{Q}\right)\right), \quad \mathbf{Q}^T\mathbf{Q} = \mathbf{I} \quad (6)$$

where Cov$(\cdot)$ denotes solving the covariance matrix, $\mathbf{I}$ stands for unit matrix.

The first principal component $\boldsymbol{\rho}_1 \in \mathbb{R}^{H\times W}$ of X is obtained by PCA as a standard segmentation pattern for ERS, the superpixel segmentation pattern $\mathbf{P}_E \in \mathbb{R}^{H\times W}$

$$\mathbf{P}_E = \bigcup_i^S \mathrm{H}_i,\ \mathrm{H}_i \cap \mathrm{H}_i = \emptyset, i \neq j \quad (7)$$

where H_i is the ith superpixel, S denotes the number of superpixels. X is divided into multiple disjoint superpixels according to $\mathbf{P}_E$, and each superpixel learns its own PCA

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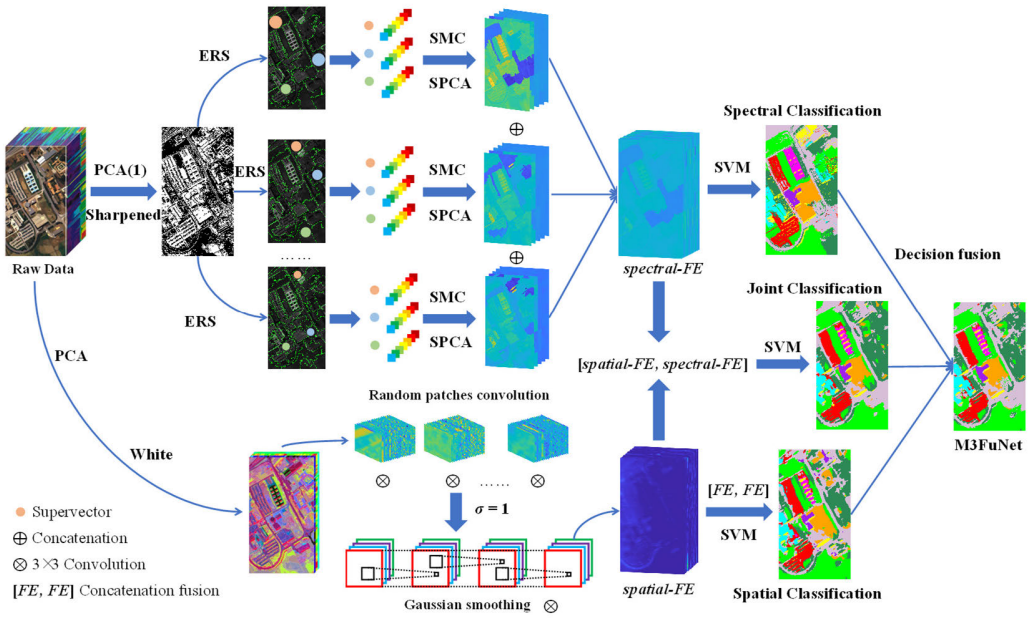


Fig. 3. Architecture of the proposed M^{3}FuNet. The top part is the MSMC algorithm. The bottom part is the MRCD algorithm. Finally, concatenation fusion and decision fusion are carried out.

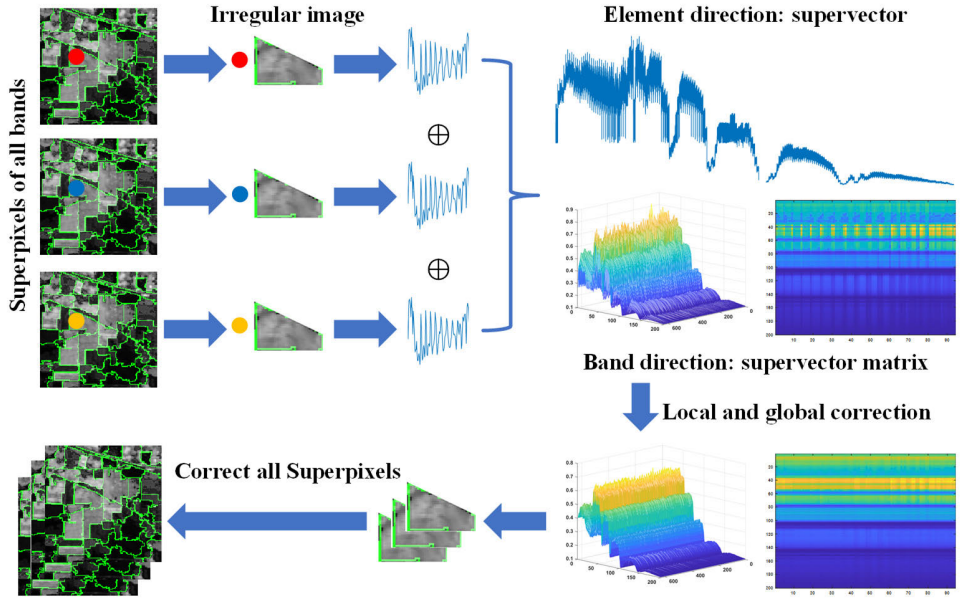


Fig. 4. Flow of SMC.

projection matrix with the same target dimension b to get $\chi_{\text{spca}} \in \mathbb{R}^{H\times W\times b}$. SuperPCA, however, cannot avoid the effect of SvBD, resulting in extremely high intraclass variance.

B. Supervector Matrix Correction

In order to alleviate the SvBD of supervector, this article proposes a SMC method, and the algorithm flow is shown in Fig. 4. Since all the bands in the same superpixel of the χ have the same number of pixels, the pixels of each band can be stretched into 1-D vector (supervector block) with the same length, and all the supervector blocks are spliced into a matrix along the band direction, this matrix is called the "supervector matrix $\mathbf{M}_s$."

A superpixel corresponds to a supervector and a supervector matrix, so defining the label of a superpixel as s, the corresponding supervector to be $\mathbf{y}_s \in \mathbb{R}^{1\times L_s}(L_s = B \times N_s)$, and the supervector matrix to be $\mathbf{M}_s \in \mathbb{R}^{B\times N_s}$, the N_s denotes the number of pixels contributed by a band (supervector block) belonging to this superpixel. The supervector block of a band is obtained by the corresponding element index from the label s

$$I_s = \text{ind}(\mathbf{P}_E = s),\ I_s \in \mathbb{R}^{1\times N_s} \tag{8}$$

$$\mathcal{U}_i = \mathbf{P}_i(I_S),\ \mathcal{U}_i \in \Re^{N_S} \tag{9}$$

$$\mathbf{y}_i = \text{fla}(\mathcal{U}_i),\ \mathbf{y}_i \in \mathbb{R}^{1\times N_S} \tag{10}$$

where $\text{ind}(\mathbf{A} = a)$ denotes the index of the element in matrix $\mathbf{A}$ equal to a, I_s denotes the array of element indexes

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belonging to the s superpixel, $\Re^a$ denotes a 2-D irregular image field with element number a, U_i denotes the irregular image corresponding to the s superpixel in $\mathbf{P}_i$, fla(·) denotes the vector expansion function, $\mathbf{y}_i$ denotes supervector block corresponding to the s superpixel in $\mathbf{P}_i$.

Cascading $\mathbf{y}_i$ along the element number direction and the band number direction to get $\mathbf{y}_s$ and $\mathbf{M}_s$

$$\mathbf{y}_s = \left[\mathbf{y}_1, \mathbf{y}_2 \cdots \mathbf{y}_B\right] \in \mathbb{R}^{1\times L_s} \tag{11}$$

$$\mathbf{M}_s = \begin{bmatrix} \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_B \end{bmatrix} \in \mathbb{R}^{B\times N_s}. \tag{12}$$

Supervector block outliers y_o are defined as elements that differs from the local median by more than three times of converted d within the specified window length w (w = 10 performs best on the three datasets) in the supervector block (local correction)

$$d = \partial * \text{med}(|\mathbf{y}_l - \mathbf{e} * \text{med}(\mathbf{y}_l)|) \tag{13}$$

$$y_o = \text{lin}\left(y_i^n\right), \text{if} y_i > 3 * d \tag{14}$$

where $\partial = (-1/(2)^{1/2} * \text{erf}((3/2)))$ denotes the conversion rate, erf(·) denotes find inverse error function, $\mathbf{e}$ stands for unit vector, med(·) denotes median function, $\mathbf{y}_l = (y_1, y_2, \ldots, y_w) \in \mathbb{R}^{1\times w}$ denotes the local vector in the supervector block with a window of w and a move step of 1. Removing filtered outliers y_o and populate using linear interpolation of neighboring nonoutliers $\text{lin}(y_i^n)$. The same local correction process is applied to all supervector blocks, and this process can be viewed as a projective transformation: $\mathbf{y}_i^* = \mathcal{T}(\mathbf{y}_i, w)$.

Then, an approximate mean constraint is imposed on the $\mathbf{y}_i^*$

$$\mathbf{y}_i' = \text{rep}^2 \frac{1}{2}\left(\mathbf{y}_i^* + \text{mea}\left(\mathbf{y}_i^*\right)\right) \tag{15}$$

where mea(·) denotes mean function, rep^2 denotes two repetitions of the calculation, $\mathbf{y}_i' \in \mathbb{R}^{1\times N_s}$ denotes corrected supervector blocks with reduced local intraclass variance.

To further reduce the global intraclass variance, two distributional primes (**as**, **bs**) and a deviation threshold D are used to filter the supervector blocks with large deviation (global correction)

$$as_i = \frac{1}{2}\left(\text{mea}\left(\mathbf{y}_i'\right) + \text{med}\left(\mathbf{y}_i'\right)\right), \mathbf{as} = (as_1, as_2, \cdots as_B) \in \mathbb{R}^{1\times B} \tag{16}$$

$$v = \left|\frac{1}{2}\text{med}\left(\left|\mathbf{as} - \text{med}(\mathbf{as})\right|\right)\right| \tag{17}$$

$$\mathbf{bs} = \mathcal{T}(\mathbf{as}, w), \mathbf{bs} = (bs_1, b\ s_2, \cdots bs_B) \in \mathbb{R}^{1\times B} \tag{18}$$

$$\mathbf{y}_i' = \mathbf{e} * bs_i., \text{if } |as_i - bs_i| > v \tag{19}$$

where **as** reflects the distributional prime of each supervector block, **bs** reflects the spectral law of the overall supervector in the ideal smooth case, v is block deviation threshold. The process simply involves replacing itself with the ideal spectral value of a deviation supervector block. Equations (14)–(20) above complete the local and global correction of the supervector matrix $\mathbf{M}_s$.

The same correction is applied to the corresponding supervector matrices of all superpixels, and the corrected supervector matrices are used to restore the 3-D cube structure $\chi_{\text{smc}} \in \mathbb{R}^{H\times W\times B}$.

Algorithm 1 Proposed MSMC spectral feature extraction method

Input: Raw data $\boldsymbol{\chi} = [\mathrm{P}_1, \mathrm{P}_2, \cdots \mathrm{P}_B] \in \mathbb{R}^{H\times W\times B}$, ERS segmentation scale z, superpixel retention scale sc and preserve principal component dimensions b_1, i denotes data in i-th band.
Initialization: Calculate ρ_1^*, P_E^* and $\mathfrak{h}$ using (5)–(7) and (21)–(23).
1: while $j < sc + 1$ do
2: for $s = 1$ *to* $\mathfrak{h}(t)$ do
3: Calculate y_i, y_s and M_s of superpixel labeled s using (8)–(13)
4: Correct y_i and update y_i to y_i' using (14)–(16)
5: Correct y_i' using (17)–(20)
6: Restore irregular superpixel subregion $\mathcal{U}_i$.
7: Apply SPCA ($\mathbf{b_1} = 5$) for the $\mathcal{U} = [\mathcal{U}_1, \mathcal{U}_2, \cdots \mathcal{U}_B] \in \mathfrak{R}^{\mathbf{N_s}\times B}$.
8: end for
9: Restore all $\mathcal{U}$ to the 3D structure χ_{smc_j}.
10: Cascade all χ_{smc_j} to $\chi_{spectral} = [\chi_{smc_1}, \chi_{smc_2}, \ldots, \chi_{smc_{sc}}]$ using (24)
11: end while
Output: $\chi_{spectral} \in \mathbb{R}^{\mathbf{H}\times\mathbf{W}\times\mathbf{B_1}}$.

C. Multiscale Supervector Matrix Correction

The flow of MSMC is shown in Fig. 5, and the MSMC algorithm is described in Algorithm 1. An unsharp masking filter (USM) [34] is used for $\boldsymbol{\rho}_1$ enhancement before superpixel segmentation. The sharpening formula can be summarized as follows:

$$\Delta = \boldsymbol{\rho}_1(x, y) - \mathbf{G_1}(x, y) \tag{20}$$

$$\boldsymbol{\rho}_1^*(x, y) = \begin{cases} \boldsymbol{\rho}_1(x, y) + \theta * \Delta, & |\Delta| > t \\ \boldsymbol{\rho}_1(x, y), & \text{otherwise} \end{cases} \tag{21}$$

where $\boldsymbol{\rho}_1^*$ is the sharpened first principal component, (x, y) represent the horizontal and vertical coordinates of the pixel in the image, $\mathbf{G_1}$ denotes the Gaussian image of $\boldsymbol{\rho}_1$, t denotes sharpening threshold, θ adjusts the decision weight of Δ. Then the ERS segmentation pattern $\mathbf{P}_E^*$ is obtained by $\boldsymbol{\rho}_1^*$ according to (7).

The multiscale superpixel segmentation can be determined by a set of approximately smooth curve values to divide the $\mathbf{P}_E^*$ into $2z + 1$ scales

$$\mathfrak{h} = \varphi * \left(\sqrt{2}\right)^{(-z,z)} \tag{22}$$

where $\mathfrak{h} \in \mathbb{R}^{1\times(2z+1)}$ denotes the number of superpixels at all scales, φ denotes the initial number of superpixels (set $\varphi = 100$ empirically), z denotes the range of superpixels

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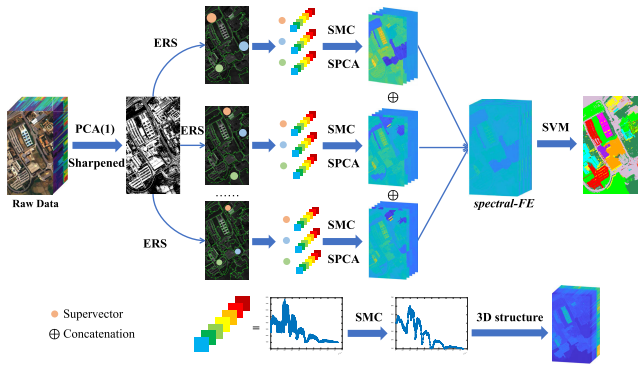


Fig. 5. Spectral classification with the proposed MSMC.

($z = 4$ to have a discussion within the nine scales). Based on these $2z + 1$ scales, SMC is applied to the raw HSI data X to obtain $2z + 1$ corrected features χ_{smc}. To further reduce the computational complexity, these $2z + 1$ corrected features were downscaled using SuperPCA with the target dimension fixed at $b_1 = 5$.

Cascading the first sc ($\text{sc} \leq 2z + 1$) features of the above $2z + 1$ corrected features $\tilde{\chi}_{\text{smc}} \in \mathbb{R}^{H\times W\times b_1}$ in order from the smallest to the largest number of superpixel scales

$$\chi_{\text{spectral}} = \left[\tilde{\chi}_{\text{smc}}1, \tilde{\chi}_{\text{smc}}2, \ldots, \tilde{\chi}_{\text{smc}}\text{sc}\right] \in \mathbb{R}^{H\times W\times B_1}$$
$$B_1 = b_1 \times \text{sc}. \tag{23}$$

The obtained spectral features χ_{spectral} are input into the support vector machine (SVM) [35] to obtain the spectral decision $R_1 \in \mathbb{R}^{H\times W}$.

IV. Multiscale Random Convolution Dispersion

A. Random Patches Convolution

To extract the deep spatial features, random patch convolution is an efficient way. Projective dimension subspace $\mathbf{X}_k \in \mathbb{R}^{L\times k}(k < B)$ of the HSI are obtained by PCA and then whitened to get $\mathbf{X}_w \in \mathbb{R}^{H\times W\times b}$

$$\mathbf{X}_w = \mathbf{X}_k\mathbf{V}_k\begin{bmatrix}\frac{1}{\sqrt{\lambda_1}} & & \\ & \ddots & \\ & & \frac{1}{\sqrt{\lambda_k}}\end{bmatrix}\mathbf{V}_k^T,\ \mathbf{U}_k = \begin{bmatrix}\lambda_1 & & \\ & \ddots & \\ & & \lambda_k\end{bmatrix}. \tag{24}$$

Denoting $\mathbf{H}_k$ is the covariance matrix of $\mathbf{X}_k$, the left singular value vector $\mathbf{V}_k$ and the singular value matrix $\mathbf{U}_k$ of $\mathbf{H}_k$ are obtained using singular value decomposition (SVD). Reconstructing $\mathbf{X}_k$ into a 3-D structure $\chi_w \in \mathbb{R}^{H\times W\times k}$, so that χ_w has similar variance across bands.

HSI is randomly divided into r patches $\text{cov}_i \in \mathbb{R}^{h\times w}$ with the same length and width as the convolution kernel to convolve $\chi_w = [\mathbf{P}_{w1}, \mathbf{P}_{w2}, \ldots, \mathbf{P}_{wk}]$, and the window move step is fixed to 1. The jth random convolution band $\mathbf{P}_{rj}$

$$\mathbf{P}_{rj} = \sum_{i=1}^{r}\mathbf{P}_{wj} \otimes \text{cov}_i \tag{25}$$

where $\otimes$ denotes the 2-D convolution operation, applying random convolution to each band of χ_w to get the random convolution features $\chi_r = [\mathbf{P}_{r1}, \mathbf{P}_{r2}, \ldots, \mathbf{P}_{rk}] \in \mathbb{R}^{H\times W\times k}$. Because of the random nature of the RPC kernels, differences in the generated convolution kernels can lead to instability classification result.

Algorithm 2 Proposed MRCD spatial feature extraction method

Input: Raw data $\boldsymbol{\chi} = [\mathrm{P}_1, \mathrm{P}_2, \cdots \mathrm{P}_B] \in \mathbb{R}^{H\times W\times B}$ and Gaussian weight parameter $\boldsymbol{\sigma}$.
Initialization: Calculate PCA-Whiten data $\boldsymbol{\chi_w}$ using (6) and (7). $\mathbf{r}$ random patches convolution kernel $\boldsymbol{cov_i}$.
1: for $\boldsymbol{j = 1\ to\ c - lay}$ do
2: Calculate P_{r_j} using (25) and (26)
3: for $\boldsymbol{h = 1\ to\ g - lay}$ do
4: Calculate P_{g_i} using (26)–(28).
5: end for
6: Cascade all P_{g_i} to $\boldsymbol{\chi}_{g_j} = [\mathrm{P}_{g_1}, \mathrm{P}_{g_2}, \ldots, \mathrm{P}_{g_k}] \in \mathbb{R}^{H\times W\times k}$.
7: end for
8: Cascade all $\boldsymbol{\chi}_g j$ to $\boldsymbol{\chi}_{spatial} = [\boldsymbol{\chi}_{g_1}, \boldsymbol{\chi}_{g_2}, \ldots, \boldsymbol{\chi}_{g_{(\text{c-lay})}}]$ using (29).
Output: $\boldsymbol{\chi}_{spatial} \in \mathbb{R}^{\mathrm{H}\times\mathrm{W}\times\mathrm{B}_2}$.

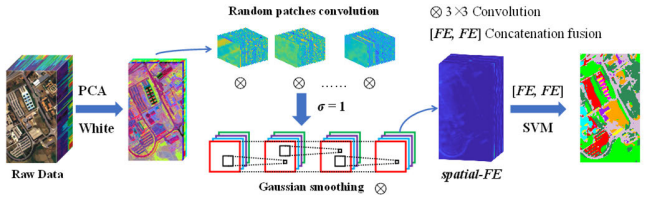


Fig. 6. Spatial classification with the proposed MRCD.

B. Multiscale Random Convolution Dispersion

In order to improve the stability of random convolution features χ_r, this article introduces Gaussian smoothing to numerically disperse the χ_r. Gaussian smoothing can reduce the change gradient of spatial features and the edge interference of heterogeneous features, connect the contextual information better. The flow of MRCD is shown in Fig. 6, and the MRCD algorithm is described in Algorithm 2. Setting the radius to be 1 (3×3 neighborhood), that is to take the target center point h and the surrounding eight points. According to the specified Gaussian weight σ ($\sigma = 1$ performs best on the three datasets), the total weight $\mathbf{W}_t$ of these nine points

$$\mathbf{W}_t(x, y) = \frac{\exp\left(\frac{-(x^2+y^2)}{2\sigma^2}\right)}{2\pi\sigma^2} \tag{26}$$

then calculate the weighted average $\mathbf{W}_a$ of these nine points

$$\mathbf{W}_a(i, j) = \frac{\mathbf{W}_t(i, j)}{\sum_{i=1}^{9}\sum_{j=1}^{9}\mathbf{W}_t(i, j)}. \tag{27}$$

$\mathbf{W}_a$ is used as a Gaussian convolution kernel with a symmetric boundary and a move step of 1 to update the eigenvalue of the center point h

$$\mathbf{P}_{gi} = \mathbf{P}_{wi} \otimes \mathbf{W}_a. \tag{28}$$

This process of computing the Gaussian convolution kernel while updating the center point h in Fig. 7, the

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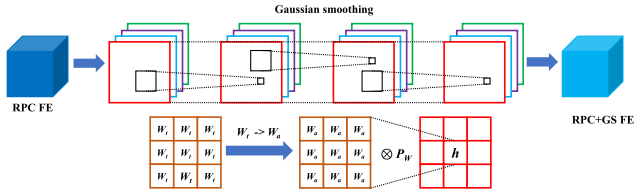


Fig. 7. Flow of Gaussian smoothing.

TABLE I
INFORMATION ABOUT DATASETS

Related information	Dataset		
	Indian pines	Pavia University	Salinas
Size (pixels)	145 × 145	610 × 340	512 × 217
Bands	200	103	224
Spatial-res (m)	20	1.3	3.7
Spectral-wave (nm)	400–2500	430–860	400-2500
Sensor	AVIRIS	ROSIS	AVIRIS
Class	16	9	16

random convolution Gaussian smoothing features are $\chi_g = [\mathbf{P}_{g1}, \mathbf{P}_{g2}, \ldots, \mathbf{P}_{gk}] \in \mathbb{R}^{H\times W\times k}$.

Taking the above random convolution Gaussian smoothing as an individual convolution layer, the number of joint convolution layers is $c-$lay, the number of Gaussian smoothing is $g-$lay. $c-$lay and $g-$lay are taken as two independent parameters, respectively; the parameter values will be discussed in the experiments section. The shallow features are cascaded and with the deep features so that the spatial features obtained have the advantages of multiscale

$$\chi_{\text{spatial}} = \left[\chi_{g1}, \chi_{g2}, \ldots, \chi_{g(c-\text{lay})}\right] \in \mathbb{R}^{H\times W\times B_2}$$
$$B_2 = k \times (c - \text{lay}). \tag{29}$$

Finally, the obtained spatial features χ_{spatial} are input into SVM to obtain the spatial decision $R_1 \in \mathbb{R}^{H\times W}$.

V. EXPERIMENTS

Three commonly used HSI datasets, including Indian Pines, Pavia University, and Salinas are used to evaluate the effectiveness of the proposed MSMC, MRCD, M^{3}FuNet and other HSI classification methods, including SVM [35], SuperPCA [12], RPNet [24], TensorSSA [27], S^3-ULDA [28], morphFormer [29], SSTN [30] and MSSFP [31]. We also analyze the effect of two important parameters ($g-$ lay of Gaussian smoothing and $c-$ lay of random convolution) on the classification performance of M^{3}FuNet.

In order to verify the effectiveness of MSMC in mitigating the SvBD and MRCD in mitigating the instability of the random convolution features, the experiments also separately analyze the MSMC and MRCD. Table I lists the relevant information of the three datasets, all experiments follow a uniform number of training-testing division. Tables II–IV list training and testing set division of the three datasets, and Figs. 8–10 demonstrate the false-color map and ground-truth map of the three datasets. All experiments were set at the same training confident level.

Three indices [36], [37], [38] are used to evaluate experimental results: overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa). The test of SVM, SuperPCA, RPNet, TensorSSA, S^3-ULDA, MSMC, MRCD, and M^{3}FuNet are all based on Windows 10, MATLAB2021a, i5-3470 CPU, and 16 GB of memory RAM (graphics processing unit (GPU) acceleration is not used). The test of morphFormer, SSTN, and MSSFP are all based on Windows 10, Pytorch1.9 framework in PyCharm, i5-12490F CPU, NVIDIA GeForce RTX 3060Ti GPU (8 GB) and 16 GB memory RAM. The experiments were repeated 30 times.

TABLE II
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE INDIAN PINES DATASET

Class	Class Name	Train	Test	Total
1	Alfalfa	5	41	46
2	Corn-notill	143	1285	1428
3	Corn-mintill	83	747	830
4	Corn	24	213	237
5	Grass-pasture	48	435	483
6	Grass-trees	73	657	730
7	Grass-pasture-mowed	3	25	28
8	Hay-windrowed	48	430	478
9	Oats	2	18	20
10	Soybean-notill	97	875	972
11	Soybean-mintill	246	2209	2455
12	Soybean-clean	59	534	593
13	Wheat	21	184	205
14	Woods	127	1138	1265
15	Buildings	39	347	386
16	Stone-Steel-Towers	10	83	93
Total	-	1028	9221	10249

TABLE III
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE PAVIA UNIVERSITY DATASET

Class	Class Name	Train	Test	Total
1	Asphaltl	67	6564	6631
2	Meadows	187	18462	18649
3	Gravel	21	2078	2099
4	Tress	31	3033	3064
5	Metal sheets	14	1331	1345
6	Bare Soil	51	4978	5029
7	Bitumen	14	1316	1330
8	Bricks	37	3645	3682
9	Shadows	10	937	947
Total	-	432	42344	42776

TABLE IV
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE SALINAS DATASET

Class	Class Name	Train	Test	Total
1	Brocoli_green_weeds_1	21	1988	2009
2	Brocoli_green_weeds_2	38	3688	3726
3	Fallow	20	1956	1976
4	Fallow_rough_plow	14	1380	1394
5	Fallow_smooth	27	2651	2678
6	Stubble	40	3919	3959
7	Celery	36	3543	3579
8	Grapes_untrained	113	11158	11271
9	Soil_vinyard_develop	63	6140	6203
10	Corn_senesced_green_weeds	33	3245	3278
11	Lettuce_romaine_4wk	11	1057	1068
12	Lettuce_romaine_5wk	20	1907	1927
13	Lettuce_romaine_6wk	10	906	916
14	Lettuce_romaine_7wk	11	1059	1070
15	Vinyard_untrained	73	7195	7268
16	Vinyard_vertical_trellis	19	1788	1807
Total	-	549	53580	54129

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TABLE V
CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR INDIAN PINES

Class	Baseline			State-of-the-art					Proposed			
	SVM	SuperPCA	RPNet	TensorSSA	S^3-ULDA	morphFormer	SSTN	MSSFP	MRCD	MSMC	M^2Fu	M^3FuNet
1	24.39	85.37	95.12	87.80	**97.56**	90.24	92.68	**97.56**	**97.56**	**97.56**	**97.56**	**97.56**
2	78.75	91.28	98.52	94.09	96.50	94.71	94.40	94.79	**99.46**	96.03	98.37	98.68
3	71.49	92.24	98.93	94.11	97.05	77.64	98.13	95.58	98.53	96.25	**100**	**100**
4	49.77	62.44	**100**	95.77	91.08	99.53	99.06	**100**	**100**	98.12	99.53	99.53
5	89.20	86.21	95.86	99.31	96.55	95.40	**99.77**	94.25	98.85	94.48	97.24	97.24
6	94.22	98.33	**100**	97.41	**100**	98.78	99.39	99.54	98.33	99.54	99.39	99.54
7	64.00	76.00	48.00	88.00	96.00	**100**	96.00	**100**	88.00	**100**	**100**	**100**
8	98.60	98.14	99.77	**100**	99.77	**100**	**100**	99.77	**100**	99.77	99.77	99.77
9	27.78	0	**100**	88.89	**100**	72.22	**100**	**100**	**100**	**100**	**100**	**100**
10	74.63	93.94	96.80	94.86	**98.17**	95.09	94.29	94.51	96.91	96.46	97.60	97.94
11	81.98	95.84	98.55	96.70	99.09	98.82	98.87	98.28	98.46	97.01	**100**	99.86
12	57.49	85.39	95.51	91.76	98.50	96.82	93.82	97.19	97.19	95.69	**99.63**	99.25
13	94.57	94.02	**100**	97.28	99.46	98.37	99.46	98.37	**100**	96.20	94.57	98.37
14	93.67	91.74	99.38	98.95	**99.91**	99.65	99.82	99.47	98.77	99.30	99.74	99.74
15	58.79	93.66	99.71	98.56	**100**	94.81	98.56	99.42	99.42	98.56	97.98	98.85
16	71.08	60.24	85.54	**100**	98.80	**100**	**100**	89.16	87.95	86.75	86.75	86.75
OA	80.12	91.95	98.21	96.28	98.36	95.84	97.69	97.28	98.48	97.15	98.98	**99.12**
AA	70.65	81.55	94.48	95.22	98.03	94.51	97.77	97.37	97.46	96.98	98.01	**98.32**
Kappa	77.26	90.77	97.96	95.76	98.13	95.24	97.36	96.90	98.27	96.75	98.84	**99.00**

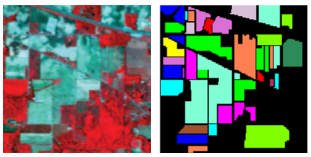


Fig. 8. False-color image and ground-truth map for the Indian Pines dataset.

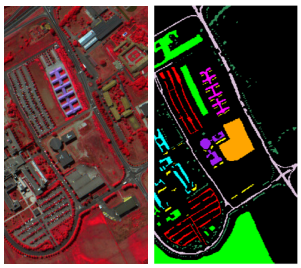


Fig. 9. False-color image and ground-truth map for the Pavia University dataset.

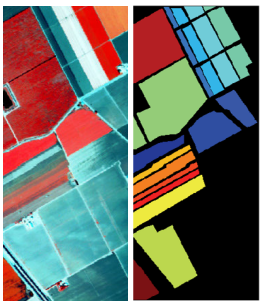


Fig. 10. False-color image and ground-truth map for the Salinas dataset.

A. Experimental Parameter Setup

1) Implementation Details: In the spectral feature extraction method MSMC, the principal component is fixed to 5 for all the datasets; in the spatial feature extraction method MRCD, the principal component is fixed to 20 for all the datasets. Specifically, in the MSMC, the scale range z is fixed to ± 4 ($2z + 1 = 9$) segmentation scales, the first seven scales (sc $= 7$) were taken for Indian Pines, the first four scales (sc $= 4$) were taken for Pavia University, the first three scales (sc $= 3$) were taken for Salinas. The number of superpixel for each of the three datasets is (round down)

$$\text{sc}_{\text{IP}} = \{\lfloor 25 \rfloor, \lfloor 35.36 \rfloor, \lfloor 50 \rfloor, \lfloor 70.71 \rfloor, \lfloor 100 \rfloor, \lfloor 141.42 \rfloor, \lfloor 200 \rfloor\}$$
$$\text{sc}_{\text{PU}} = \{\lfloor 25 \rfloor, \lfloor 35.3553 \rfloor, \lfloor 50 \rfloor, \lfloor 70.7107 \rfloor\}$$
$$\text{sc}_{\text{SA}} = \{\lfloor 25 \rfloor, \lfloor 35.3553 \rfloor, \lfloor 50 \rfloor\}.$$

2) State-of-the-Art Methods: SVM: SVM classifier is implemented by the library for SVM (LIBSVM) [39] using the radial basis function (RBF) kernel with fivefold cross validation. SuperPCA: superpixel for the three datasets is fixed to 50. RPNet: the dimensionality reduction per layer is set to 3, the size of the convolution window is 20 × 20, and convolution layers are set to 5. TensorSSA: the values of the spatial level and the embedding exposure corresponding to the three datasets are set to (49, 5), (15, 2), and (20, 3), respectively. S^3-ULDA: the number of superpixel is set to 35, the reduced feature dimension is set to 15, and the nearest neighbor is set to 15. MorphFormer: patch size of the convolutional layer set to $11 \times 11 \times B$ and epochs are 500 for three datasets. SSTN: batch size is 32, and epochs are 100 for three datasets. MSSFP: batch size is 64, and epochs are 300 for three datasets.

B. Quantitative Experimental Results of M^3FuNet

This section first demonstrates the sensitivity of the M^3FuNet to the combination of two important parameters $g-$lay and $c-$lay in the MRCD method (example of OA), with the alternative values of variation being $g-$lay: {1, 2, 3, 4, 5, 6} and $c-$lay: {1, 2, 3, 4, 5, 6}. As shown in Fig. 11, the parameter $c-$lay as a factor dominating the performance in Indian Pines and Pavia University, the performance tends to stabilize after three layers, and the $g-$lay value has a greater degree of influence when the $c-$lay value is low, and $g-$lay is almost dominant when the $c-$lay value is large. The influence ability of the two parameters is almost equal in Salinas. The optimal parameter combinations ($g-$lay, $c-$lay) for three datasets are proved to be (5, 2), (3, 3), and (4, 3), respectively.

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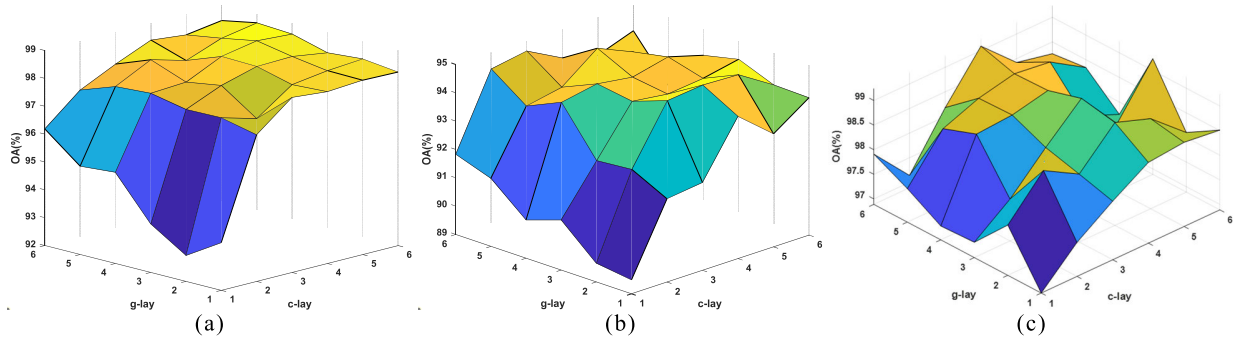


Fig. 11. Parameter tuning of $g-$lay and $c-$lay for the proposed M^{3}FuNet using the three experimental datasets. (a) Indian Pines. (b) Pavia University. (c) Salinas.

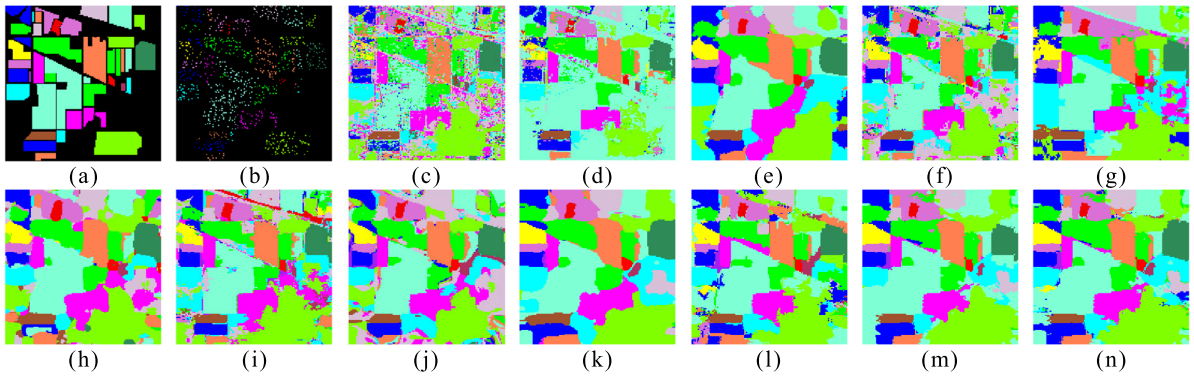


Fig. 12. Classification maps obtained by different methods on the Indian Pines dataset. (a) Ground-truth. (b) Training samples. (c) SVM. (d) SuperPCA. (e) RPNet. (f) TensorSSA. (g) S^3-ULDA. (h) morphFormer. (i) SSTN. (j) MSSFP. (k) MRCD. (l) MSMC. (m) M^{2}Fu. (n) M^{3}FuNet.

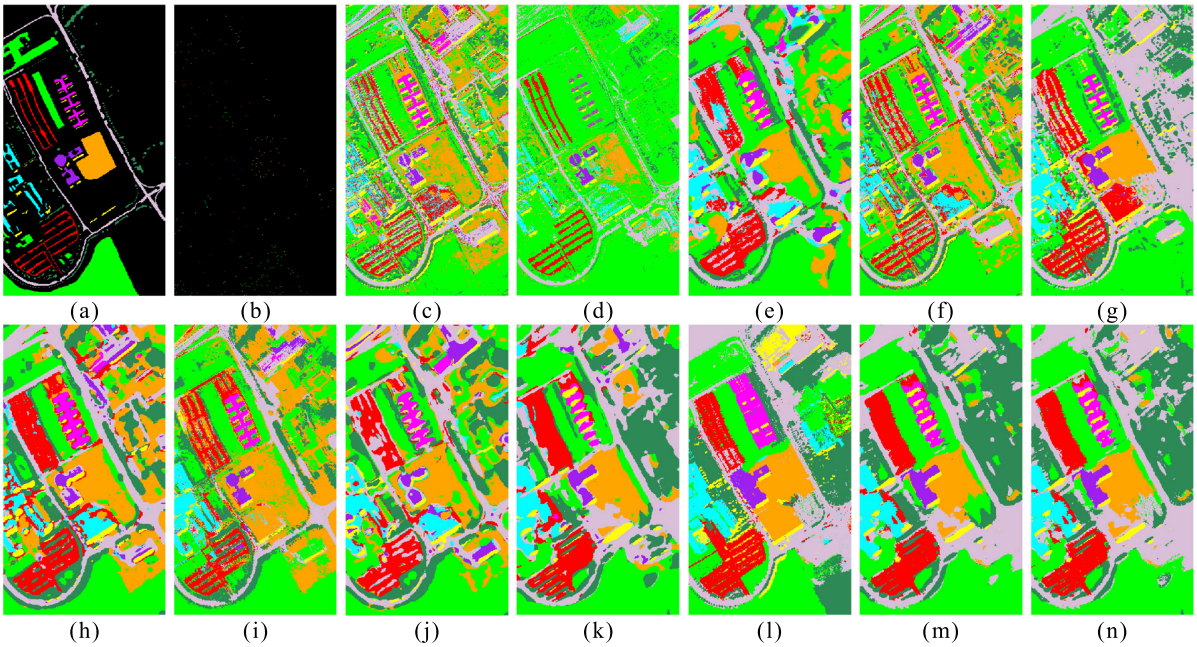


Fig. 13. Classification maps obtained by different methods on the Pavia University dataset. (a) Ground-truth. (b) Training samples. (c) SVM. (d) SuperPCA. (e) RPNet. (f) TensorSSA. (g) S^3-ULDA. (h) morphFormer. (i) SSTN. (j) MSSFP. (k) MRCD. (l) MSMC. (m) M^{2}Fu. (n) M^{3}FuNet.

For a more intuitive comparison of the proposed methods for ablation, the spatial–spectral joint feature classification result is M^{2}Fu (MSMC and MRCD feature concatenation fusion). Tables V–VII show the OA, AA, and Kappa indices for Indian Pines, Pavia University, and Salinas in different methods, respectively.

SVM, SuperPCA, and RPNet are the baseline methods with different ideas for extracting features. SVM mainly analyses pixel-level spectral information, lacks consideration of spatial information, and the overall classification performance is poor. SuperPCA is a dimensionality reduction method based on superpixel guidance, which can form some spectral-spatial

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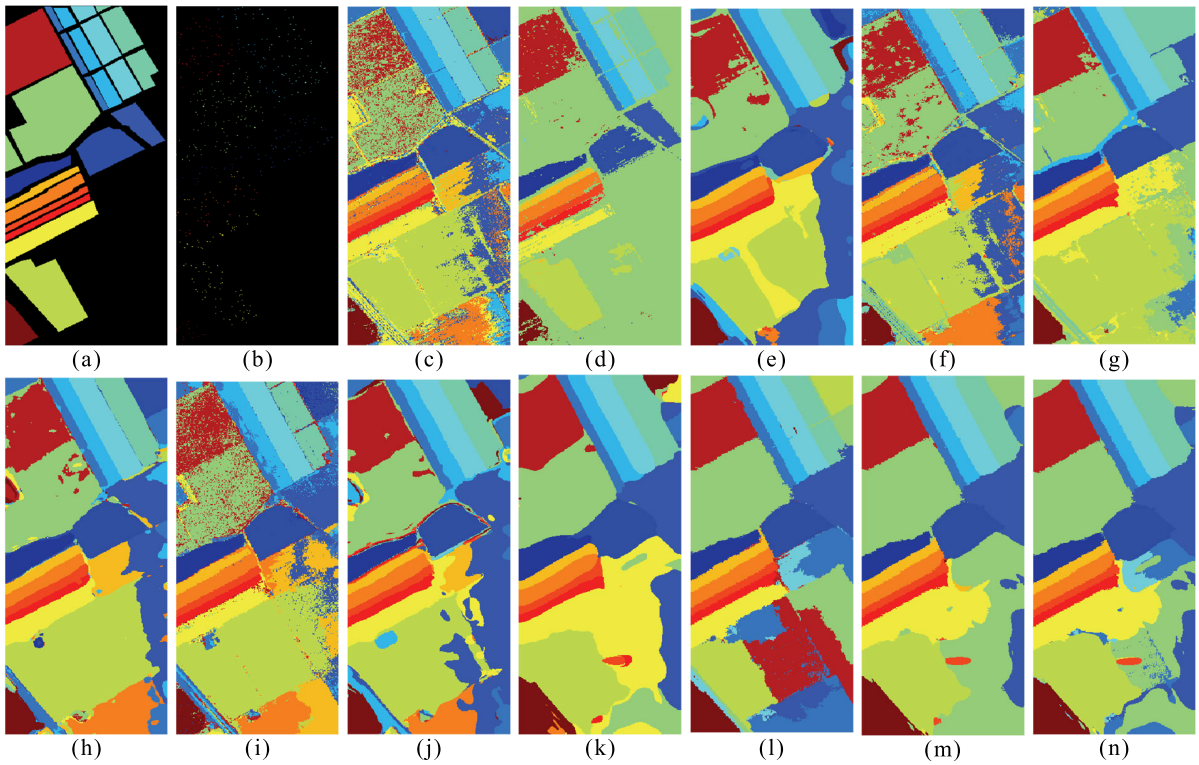


Fig. 14. Classification maps obtained by different methods on the Salinas dataset. (a) Ground-truth. (b) Training samples. (c) SVM. (d) SuperPCA. (e) RPNet. (f) TensorSSA. (g) S^3-ULDA. (h) morphFormer. (i) SSTN. (j) MSSFP. (k) MRCD. (l) MSMC. (m) M^2Fu. (n) M^3FuNet.

TABLE VI

CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR PAVIA UNIVERSITY

Class	Baseline			State-of-the-art					Proposed			
	SVM	SuperPCA	RPNet	TensorSSA	S^3-ULDA	morphFormer	SSTN	MSSFP	MRCD	MSMC	M^2Fu	M^3FuNet
1	86.56	88.44	96.05	97.26	95.53	93.34	84.98	92.03	94.39	97.65	98.22	**99.24**
2	95.24	97.77	98.02	99.52	97.68	99.89	99.88	99.37	99.26	**99.96**	**99.96**	**99.96**
3	62.32	68.00	74.30	90.04	96.50	70.79	96.58	79.16	86.96	**99.13**	97.74	98.60
4	89.25	59.02	85.03	90.37	91.83	**95.02**	89.75	91.13	82.39	71.22	82.10	81.17
5	98.95	49.59	87.68	98.95	99.92	**100**	99.62	98.12	76.18	99.02	90.98	91.36
6	79.29	77.68	98.57	95.56	99.44	99.82	94.84	97.83	99.98	99.42	99.94	**100**
7	80.55	89.06	69.00	93.85	**100**	87.61	96.05	76.82	91.03	**100**	**100**	**100**
8	79.81	92.81	91.91	89.90	98.50	97.92	91.82	86.39	97.31	98.82	99.09	**99.18**
9	99.89	85.81	64.25	96.58	**100**	85.59	59.45	69.26	69.48	62.22	74.07	71.93
OA	88.41	87.25	93.19	96.50	97.34	96.23	94.37	93.94	94.97	96.48	97.37	**97.48**
AA	85.76	78.69	84.98	94.67	**97.71**	92.22	90.33	87.79	88.55	91.94	93.57	93.49
Kappa	84.58	82.99	90.94	95.34	96.48	95.00	92.51	91.95	93.31	95.30	96.50	**96.65**

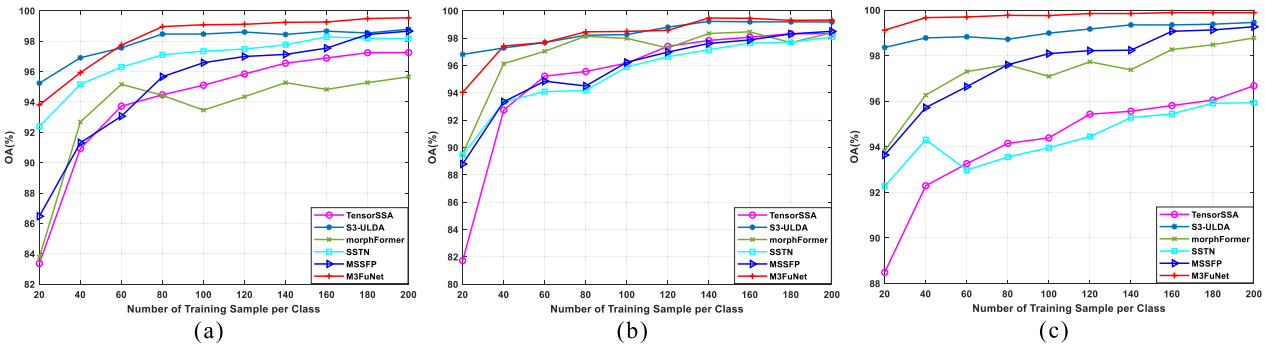


Fig. 15. Overall classification accuracy of different methods versus different numbers of training samples per class for the three experimental datasets. (a) Indian Pines. (b) Pavia University. (c) Salinas.

dependence, but the larger intraclass variance leads to poor classification accuracy. RPNet is capable of extracting deep features. The results of repeated experiments show that the average value of accuracy can reach the level of advanced methods, but the instability of the random convolution features leads to a large variance in accuracy. TensorSSA and

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TABLE VII

CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR SALINAS

Class	Baseline			State-of-the-art					Proposed			
	SVM	SuperPCA	RPNet	TensorSSA	S^3-ULDA	morphFormer	SSTN	MSSFP	MRCD	MSMC	M^{2}Fu	M^{3}FuNet
1	97.03	97.64	98.19	98.04	**100**	99.95	99.70	99.65	97.33	99.75	**100**	**100**
2	99.65	90.02	98.51	99.19	**100**	**100**	99.40	96.99	99.73	**100**	**100**	**100**
3	89.62	97.34	95.81	99.28	**100**	99.03	99.64	96.93	**100**	**100**	**100**	**100**
4	98.55	88.19	**100**	84.49	99.42	97.97	99.49	99.13	**100**	99.93	**100**	**100**
5	97.40	97.43	98.04	96.83	99.06	**99.25**	97.66	98.04	98.87	99.17	99.13	99.13
6	99.95	89.33	98.93	99.77	**99.97**	**99.97**	99.74	99.90	99.57	99.34	99.90	99.90
7	99.29	94.64	99.41	99.18	99.92	**100**	98.67	99.52	99.69	99.27	99.63	99.66
8	75.00	99.11	93.49	88.68	98.72	93.58	90.64	94.87	97.18	99.95	**99.99**	**99.99**
9	99.28	98.36	99.35	99.38	99.93	99.90	99.92	99.71	**100**	99.95	**100**	**100**
10	87.46	77.87	**99.69**	91.40	96.70	96.67	94.67	98.89	96.83	97.47	99.20	99.20
11	86.85	79.09	97.26	92.53	94.99	99.53	99.23	98.49	98.96	**100**	**100**	**100**
12	93.45	92.66	97.69	**100**	99.90	99.74	99.95	**100**	98.48	**100**	99.79	99.79
13	97.79	94.92	97.35	98.57	98.23	**100**	99.21	99.12	99.67	97.68	97.68	97.68
14	91.31	75.45	96.79	89.05	**98.30**	95.56	96.25	98.39	98.02	96.41	98.02	97.54
15	56.11	90.60	82.54	80.07	94.82	93.58	71.73	95.25	99.62	99.33	**99.74**	**99.74**
16	94.46	81.66	**100**	91.83	99.05	99.22	95.95	98.32	**100**	**100**	**100**	**100**
OA	86.44	92.73	95.43	92.93	98.56	97.33	93.35	97.57	98.82	99.48	**99.75**	**99.75**
AA	91.45	90.27	97.07	94.27	98.69	98.37	96.37	98.32	99.00	99.27	**99.57**	99.54
Kappa	84.90	91.82	94.90	92,13	98.40	97.03	92.58	97.30	98.69	99.42	**99.73**	99.72

TABLE VIII

MIOU (%) OF DIFFERENT METHODS FOR THE THREE EXPERIMENTAL DATASET

-	TensorSSA	S^3-ULDA	morphFormer	SSTN	MSSFP	M^{3}FuNet
Indian Pines	91.55	95.01	90.29	95.75	91.62	**96.08**
Pavia University	91.85	**94.86**	87.26	84.39	81.79	90.59
Salinas	89.70	97.39	96.47	93.81	96.45	**99.06**

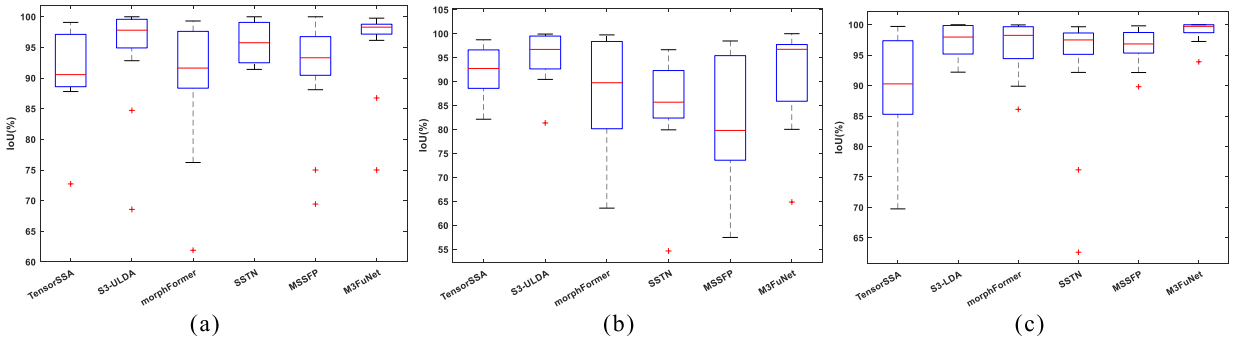


Fig. 16. IoU of different methods for the three experimental datasets. (a) Indian Pines. (b) Pavia University. (c) Salinas.

S^3-ULDA are spectral-spatial joint feature extraction methods based on superpixel guidance. TensorSSA is able to extract 3-D tensor features and is able to reach a high level in the Indian Pines and Pavia University, but performs mediocrely in the Salinas. S^3-ULDA can take the spectral-spatial joint information from both global and local perspectives, and the extracted features have a strong representation ability, obtaining high accuracy on all three datasets. morphFormer, SSTN, and MSSFP are deep learning methods based on the transformer architecture, using methods such as filter transforms and attention mechanisms to extract deep semantic features through a multilayer network structure; the deep semantic features contain rich spectral-spatial joint information and can achieve high classification accuracy on all three datasets.

MSMC and MRCD have similar results on the three datasets, indicating that the extracted spectral and spatial features have similar representation abilities. On all three datasets, the three indices and most of the single-class accuracy of the M^{2}Fu are higher than MSMC and MRCD, which validates that the fusion of such features (with similar representation ability) can mitigate the SCFC. But there are also poorly performing categories. In Classes 13 and 15 in the Indian Pines, the joint features (M^{2}Fu) have the lowest accuracy.

TABLE IX

EXECUTION TIME (IN SECONDS) OF DIFFERENT METHODS

Classifier	Indian Pines	Pavia University	Salinas
SVM	2.66	1.11	6.65
SuperPCA	12.27	6.92	14.56
RPNet	3.66	7.41	9.83
TensorSSA	66.89	154.42	199.61
S^3-ULDA	15.43	894.81	363.02
morphFormer	4120.98	9948.38	7874.55
SSTN	3835.26	8217.93	5928.06
MSSFP	284.82	373.94	363.86
MRCD	5.50	50.21	13.90
MSMC	62.01	112.28	60.29
M^{3}FuNet	68.89	165.74	78.31

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TABLE X

COMPARISON OF CLASSIFICATION ACCURACY BETWEEN ORIGINAL AND CORRECTED DATA ON THE INDIAN PINES

-	RAW	1	2	3	4	5	6	7	8	9
OA(%)	79.89	93.70	95.00	94.82	95.41	96.19	96.77	**96.99**	96.33	96.44
AA(%)	70.76	92.39	94.36	94.46	95.14	95.45	96.25	**96.88**	96.67	96.17
Kappa(%)	77.03	92.81	94.30	94.09	94.77	95.66	96.32	**96.56**	96.16	95.94

TABLE XI

COMPARISON OF CLASSIFICATION ACCURACY BETWEEN ORIGINAL AND CORRECTED DATA ON THE PAVIA UNIVERSITY

-	RAW	1	2	3	4	5	6	7	8	9
OA(%)	87.69	**97.21**	95.90	96.50	95.43	95.50	95.82	94.94	95.06	94.11
AA(%)	85.11	**94.78**	92.88	92.33	91.59	90.75	91.17	90.71	90.97	91.96
Kappa(%)	83.63	**96.28**	94.54	95.33	93.91	94.01	94.43	93.27	93.42	92.20

TABLE XII

COMPARISON OF CLASSIFICATION ACCURACY BETWEEN ORIGINAL AND CORRECTED DATA ON THE SALINAS

-	RAW	1	2	3	4	5	6	7	8	9
OA(%)	87.99	**98.96**	**98.96**	97.84	97.83	97.39	97.25	97.13	95.77	96.62
AA(%)	92.87	98.75	**98.94**	98.65	98.65	98.39	98.33	98.31	97.65	98.12
Kappa(%)	86.63	**98.84**	**98.84**	97.60	97.58	97.09	96.94	96.80	95.28	96.24

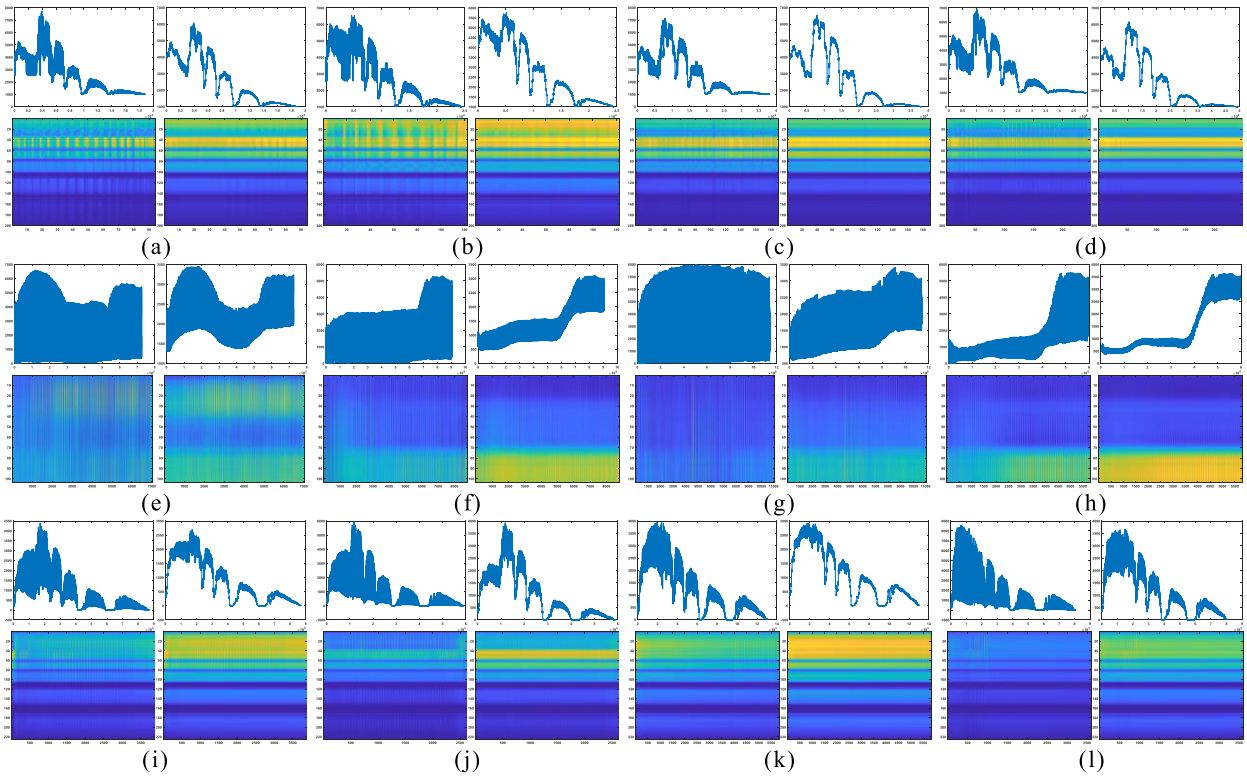


Fig. 17. Visualization of supervectors and supervector matrices corresponding to the three experimental datasets. The left side represents the original supervectors and supervector matrices, and the right side represents the corrected supervectors and supervector matrices. (a)–(d) Indian Pines. (e)–(h) Pavia University. (i)–(l) Salinas.

In Class 5 in the Pavia University, the accuracy of the spectral features is 99.02% and the accuracy of the spatial features is 76.18%, which is too large a difference in representation ability for the joint features to be sufficiently close to the better features. M^{3}FuNet in the Indian Pines has a small increase compared to M^{2}Fu, which pulls class 13 and 15 of the Indian Pines back to accuracy on par with MSMC and MRCD, further alleviates the SCFC. But class 9 of Pavia University still achieved only 91.36% accuracy. The three indices and single-class accuracy of M^{2}Fu are equal to the M^{3}FuNet in Salinas, and there is no obvious SCFC, proving that the joint features almost reach the upper limit of representation. Overall, by combining the three datasets and the ablation results (MSMC, MRCD, M^{2}Fu, and M^{3}FuNet), M^{3}FuNet can

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effectively mitigate SCFC and achieve the highest OA in the experiment.

Classification maps of the different methods were shown in Figs. 12–14. Pixel-level classifier SVM generates a lot of pretzel noise. SuperPCA loses a lot of image semantic information. TensorSSA, S^3-ULDA SSTN, and MSMC consider more spectral information and some noise contamination at the edges. RPNet, morphFormer, MSSFP, and MRCD consider more spatial information and smoother edges of the classification map. The spectral and spatial information considered in M^{2}Fu and M^{3}FuNet are more balanced and form a strong spectral-spatial dependence, which extract a complete image semantic contours for Indian Pines and Salinas, but in the Pavia University, the extracted image semantic contours is not clear due to the poor accuracy of classes 5 and 9.

In order to compare the sample sensitivity of the M^{3}FuNet and state-of-the-art (SOTA) methods, the OA with 20–200 training samples per class was tested. As shown in Fig. 15, with the training size increasing, the OA of all methods increasing and the proposed M^{3}FuNet has the best OA compared to other SOTA methods. In addition, we use each class intersection over union (IoU) to further compare the classification equilibrium of the M^{3}FuNet and SOTA methods. Fig. 16 shows the distribution and outliers of each class IoU in the three datasets; the boxes represent IoU distribution, "redline" in the boxes represents the median, and "+" represents outliers. Table VIII shows the mean IoU (mIoU) [40] of three datasets. In general, the more concentrated the IoU distribution, the fewer the outliers, and the higher the mIoU, it will be more balanced the classification performance. Fig. 16 and Table VIII illustrate that M^{3}FuNet performs optimally on the Indian Pines and Salinas and achieves advanced levels at Pavia University.

Table IX demonstrates the execution time (feature extraction to the end of classification) of all the methods; the proposed M^{3}FuNet has advantages in terms of implementation efficiency. In addition, M^{3}FuNet is performing serial experiments, MSMC and MRCD are executed sequentially, and in real application scenarios it can be executed in parallel as well, then the time spent is almost the same as MSMC.

C. Validation of the Effectiveness of MSMC for SvBD

This section evaluates the effectiveness of the SMC for SvBD in the MSMC method. The classification results of the data before correction (RAW) are compared with the classification results of the nine superpixel segmentation scales after correction (SMC), and the classifier is SVM. Tables X–XII list the classification results before and after correction. The spectral dimension is kept the same as the original dimension, and the results show that the Indian Pines perform best at the seventh scale, OA is 13%–15% above RAW, AA is 21%–25% above RAW, and Kappa is 15%–19% above RAW, the Pavia University performs best at the first scale, OA is 7%–10% above RAW, AA is 5%–9% above RAW, and Kappa is 9%–13% above RAW, the Salinas performs best on the second scale, OA is 9%–11% above RAW, AA is 5%–9% above RAW, and Kappa is 9%–12% above RAW.

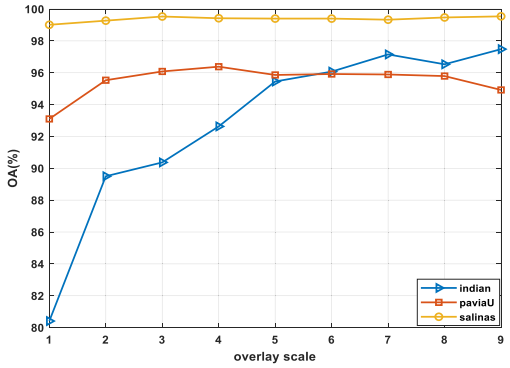


Fig. 18. Performance of concatenation fusion of low dimensional features on the three experimental datasets at different scales.

In the seventh, first and second segmentation scales with the highest classification accuracy in the Indian Pines, Pavia University, and Salinas, the (4th, 33rd, 85th, and 160th), (5th, 13th, 18th, and 22nd), and (5th, 15th, 25th, and 33rd) superpixels were selected, respectively. As shown in Fig. 17, the results of the supervectors before and after the SMC are visualized, the intraclass variance of the corrected supervectors is significantly reduced, and the supervector matrix has higher spectral similarity within bands (row data) and stronger spectral continuity between bands (column data). At lower segmentation scales, however, there may be multiple different classes of features within a superpixel, and it can be seen from the Pavia University visualization results that a portion of the supervectors do not have obvious spectral law, which indicates that there is still a need to strengthen the priori condition of superpixel segmentation.

After correcting the supervector using SMC and superpixel dimensionality reduction, the representation ability of concatenation fusion at various scales is shown in Fig. 18. One accuracy extreme point occurs at the seventh scale for the Indian Pines. Two accuracy extreme points occur at the fourth and sixth scales for Pavia University. One accuracy extreme point occurs at the third scale for the Salinas. This is also the basis for sc taking values in the three datasets in quantitative experiments.

D. Verification of Feature Space Robustness of MRCD

This section focuses on the validation of the effect of Gaussian smoothing to improve the robustness of the random patch convolution feature. As shown in Fig. 19, in 15 repeated parallel experiments, the accuracy curve after Gaussian smoothing (yes) is more stable compared to the ordinary random patch convolution (no), and the accuracy is higher in most cases. The mean values of the accuracy curves of the three datasets after Gaussian smoothing are 98.23%, 94.15%, and 98.81%, with variances of 0.035, 0.365, and 0.079, respectively, while the mean values of the accuracy curves of the ordinary random patch convolution are 97.97%, 93.01%, and 98.57%, with variances of 0.098, 0.881, and 0.148, respectively. The experimental results show that the introduction of random convolution under Gaussian smoothing can obtain more robust spatial features and higher classification accuracy.

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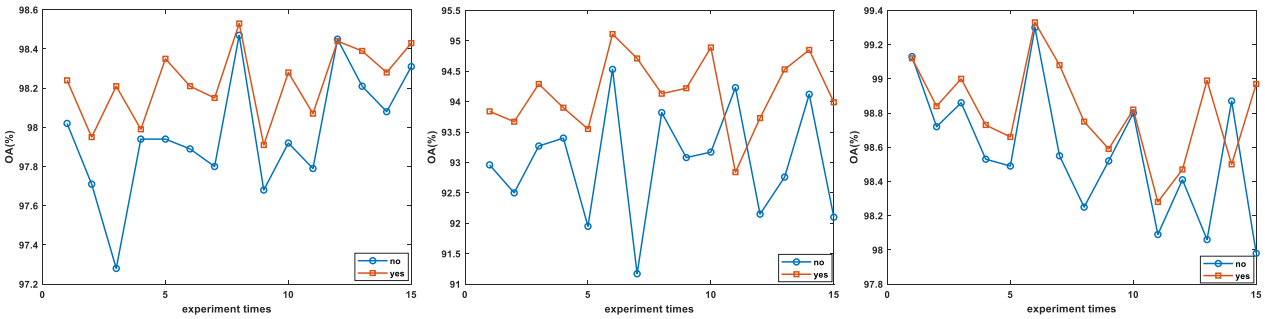


Fig. 19. Comparison of performance between, including Gaussian smoothing (curve yes) and not including Gaussian smoothing (curve no) using the three experimental datasets in multiple experiments. (a) Indian Pines. (b) Pavia University. (c) Salinas.

VI. Conclusion

In this article, an unsupervised M^{3}FuNet is developed for HSIC. M^{3}FuNet uses multivariate feature fusion (feature fusion used together with decision fusion) to improve the feature retention and achieve feature calibration, which alleviates the constraints of SCFC on classification performance. In the M^{3}FuNet, the MSMC method is used for spectral feature extraction, and the MRCD method is used for spatial feature extraction. MSMC uses the SMC algorithm to correct the supervector matrix at different segmentation scales, which alleviates the influence of SvBD. MRCD is mainly aimed at the problem of the instability of random convolution features and introduces the Gaussian smoothing to make the feature's edges smoother, which improves the robustness of random convolution features. The features extracted by MSMC and MRCD are proved to have similar representation ability after SVM learning, so that the joint features can alleviate the SCFC effectively. We conducted experiments on three commonly used HSI datasets, and the experimental results show that the proposed M^{3}FuNet has superior performance compared with some SOTA methods.

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